The Waterfront Brief

Weekly sustainment intelligence for naval shipyard leadership · Naval shipbuilding, ship repair, and fleet sustainment

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This issue covers four stories. The first asks who built each section of a new destroyer's hull and where the production records live — the answer has changed. The second covers what the Navy learned during the first Block III submarine maintenance cycle. The third looks at what the FY27 ship maintenance budget actually says versus what the past year of operations has already consumed. The fourth examines what happens to foundational trade skill when automation absorbs the work.

Labels: FACT is attributed to the named source in the same sentence; Assessment is this desk's analysis, built only on facts stated above it; Speculation is flagged as such; Unverified marks a claim this desk could not check against a primary source, with the reason given.

WATERFRONT OPS & PRODUCTION

The destroyer at your pier may have been built by six different companies, and the paperwork does not always follow

When a new Arleigh Burke-class destroyer arrives for its first scheduled availability, its hull was not built in one place. Since 2025, Huntington Ingalls Industries has been fabricating destroyers across a network of partner yards in eleven states. Roughly half of each ship's structural sections are made elsewhere and shipped to Pascagoula, Mississippi, where Ingalls welds them into the final hull.

A structural section, called a grand block in the trade, is one of the prefabricated pieces a shipyard assembles to form the hull of a ship. A typical Arleigh Burke contains about 77 of them.

FACT: according to HII's announcement at the Sea-Air-Space conference on April 20, 2026, HII planned to have 37 of those 77 sections on each destroyer built by partner yards in 2026, up from fewer than five historically. The three named partners are Trident Maritime Systems, with facilities in Michigan, Virginia, and Louisiana; Gulf Copper, based in Texas; and Keel, with facilities in Michigan and South Carolina. Together they operate across 25 locations in 11 states. FACT: per the same announcement, HII planned to outsource more than 2.5 million labor hours in 2026, a 30 percent increase from 2025. Ingalls Shipbuilding President Brian Blanchette described the model: “These major ship sections arrive ready to be erected, so we can focus our workforce on work only a major shipyard can do.”

FACT: the first partner-built sections went up on USS Thad Cochran (DDG 135) in July 2026, per HII's announcement on July 15. Gulf Copper and Eastern Shipbuilding made those first sections. For DDG 137, HII has contracted 32 sections across five different companies. For DDG 139, 37 sections are targeted for partners across another configuration.

The partner lineup changes ship to ship. The same section type that Gulf Copper built on one destroyer may be built by a different company on the next. There is no fixed assignment of section types to specific yards across the program.

FACT: on July 27, 2026, HII announced the model has expanded beyond destroyers. Eight structural sections for USS Philadelphia (LPD-32) have been awarded to two partner yards and are already in production. LPD-32 is a San Antonio-class amphibious transport dock, a large landing ship that carries Marines and their vehicles and equipment. This is the first time the distributed model has applied to the amphibious program.

FACT: on August 6, 2026, HII announced performance-based production agreements worth up to $900 million over seven years with Path Robotics and GrayMatter Robotics, under the program name HYPR, which stands for High-Yield Production Robotics. Path Robotics builds machines that weld metal automatically. GrayMatter Robotics builds machines that prepare and coat metal surfaces and inspect them. HII says the agreements extend across its carrier, submarine, destroyer, amphibious, frigate, and unmanned programs, and that GrayMatter's systems deliver up to 12 times the output of manual surface preparation with a 95 percent reduction in rework.

That establishes the scale. Now the repair side of the question.

What this means when the hull arrives at your facility

The sections met the same quality standard regardless of which partner built them. That part is not in question. The question is what documentation travels with the hull.

Production records are the paper trail that proves how a section was built: weld maps showing which joints were made in what sequence, material certifications showing where the steel came from, inspection signoffs at each stage, and quality system records showing who reviewed what and when. For a repair yard, these records answer the questions that come up the moment you open something. Which filler metal was used here? Was this weld post-heat treated? Who accepted this joint? In a single-yard build, all of that lives in one quality system under one roof. In a distributed build, each section's production records are held by whichever company made it, in that company's format, under that company's quality system.

FACT: a July 2026 RealClearDefense analysis of distributed shipbuilding reported that over 200 U.S. Navy ships already in service lack complete digital as-built models. An as-built model is the record of what was actually built and how, as distinct from the original design. The gap exists even in traditional builds; distributed fabrication makes it wider.

Non-destructive testing is the sharpest example of why those records do not simply merge when the sections arrive at Pascagoula. Non-destructive testing, or NDT, is the inspection technique that checks welds for defects without cutting them open. FACT: NAVSEA's NDT regulation, T9074-AS-GIB-010/271, states that "procedures shall not be transferred from one activity to another without the specific approval of NAVSEA." Each partner yard must establish its own written procedures and its own certified inspectors. The acceptance criteria are the same everywhere. The procedure used on any given weld, and the inspector who signed it off, are specific to whichever yard made that section. Those records stay with that yard's quality system.

FACT: a July 2026 RealClearDefense analysis of distributed shipbuilding argued that "when second- and third-tier suppliers carry parts of the build, the evidence architecture connecting them to the prime becomes the difference between defensible quality and unverifiable assumption." HII has said that partner-built sections are "fully integrated into HII's quality and material-control systems," which describes how data flows during construction. It does not specify what format of records accompanies the hull when it delivers to the fleet.

Assessment: the production records gap that already exists in traditionally-built ships is wider in a distributed build. On DDG 137, built across five companies, a repair yard asking who made a given section, what procedure covered its welds, and where those records are held will not find the answer on the hull. The problem is not that partner-built sections are lower quality. The problem is that the production history is distributed across companies that change ship to ship, and retrieving it takes knowing who to call — before the work is already scoped.

> **Bottom line:** before a new-construction DDG or LPD arrives for its first availability, contact the ship's NAVSEA program office and ask for the distributed partner list by section number, and ask where each partner's production records are held. Do this before the hull is in your dry dock.

Sources: HII Sea-Air-Space 2026, Apr 20, 2026; HII DDG 135 erection milestone, Jul 15, 2026; HII LPD-32 announcement, Jul 27, 2026; HII HYPR announcement, Aug 6, 2026; [NAVSEA T9074-AS-GIB-010/271 via Atlantis NDT]; RealClearDefense, Jul 17, 2026 {.sources}

Portsmouth completed the first Block III submarine maintenance cycle seven months over the original plan — and that number is yours to learn from

Portsmouth Naval Shipyard returned USS North Dakota to the fleet on August 29, 2026, completing the first depot-level maintenance period ever performed on a Block III Virginia-class submarine. The job ran 40 months against an original plan of 33.

The availability type is called an Extended Drydocking Selected Restricted Availability, or EDSRA: the deepest scheduled maintenance a submarine goes through, requiring the boat to leave the water entirely so the hull, propulsion systems, and internal equipment can all be reached and overhauled. FACT: USS North Dakota (SSN 784) arrived at Portsmouth on April 28, 2023 and undocked on February 26, 2026, per reporting in Army Recognition and Interesting Engineering drawing on Navy and shipyard sources. The boat completed sea trials and pier-side work by August 20, with formal fleet return on August 29. FACT: the work package was valued at approximately $347 million, per the same reporting. FACT: the availability finished ahead of its rebaselined schedule — meaning the Navy formally extended the plan during execution to reflect what the first-of-type scope actually required, and Portsmouth still beat that revised target — but the gap against the original 33-month estimate was seven months.

Block III is a distinct variant. The 11 Virginia-class boats built before USS North Dakota all share the same basic bow section: a spherical, air-backed sonar array surrounded by 12 individual vertical launch tubes for missiles. FACT: Block III replaced that entire forward section with two systems not found on earlier boats. The sonar array became a Large Aperture Bow array — horseshoe-shaped and water-backed, a different structure requiring different mounts, different access geometry, and different maintenance procedures than the spherical array it replaced. The 12 individual launch tubes became two Virginia Payload Tubes, each 87 inches in diameter and capable of holding six Tomahawk missiles, a structural change with no equivalent in the Block I/II maintenance record. FACT: eight submarines share the Block III configuration, starting with USS North Dakota, per Navy program documentation.

Portsmouth now has that baseline data. Nobody had it before.

FACT: Interesting Engineering's reporting on the delivery described the availability as establishing "a foundation for maintaining the remaining Block III boats." Portsmouth worked through the unfamiliar systems during this availability. The 7-month premium is not a performance shortfall — it is the cost of being first on a hull that looks like prior Virginias until you open the bow section.

Assessment: every maintenance facility that will touch a Block III submarine is still on the leading side of that learning curve. The structural differences in the bow section — the LAB array geometry, the Virginia Payload Tube interfaces, the access routing that fits neither Block I/II precedent — will surface during any work that touches the forward section, not only during an EDSRA. A private yard scheduling its first Block III availability should expect its scope to run differently from Block I/II experience and plan accordingly. The data Portsmouth generated during this 40-month availability is the advance notice.

Bottom line: before your first Block III Virginia-class availability — whether an EDSRA at a public yard or a Selected Restricted Availability at a private one — identify which systems in the bow section your planning is drawing on Block I/II experience for, and ask the program office for Portsmouth's Block III baseline scope documentation. Seven months of extra cost at a public yard is the number that says those two bow systems need to be on your pre-planning list.

Sources: Army Recognition, Aug 2026; Interesting Engineering, Aug 2026; DVIDS Portsmouth undocking, Feb 26, 2026; NavyCRF, Aug 31, 2026 {.sources}

The $14.3 billion ship maintenance request is the pre-Epic Fury number — the actual FY27 demand is higher

The Navy's FY27 budget asks Congress for $14.3 billion in ship depot maintenance funding, 5.5 percent more than FY26. That growth is real. What the number does not show is the gap opened by six months of sustained combat operations, and a yard that plans its FY27 capacity against the 5.5 percent growth rate alone is working from the wrong baseline.

Ship depot maintenance, account 1B4B in the Operation and Maintenance, Navy appropriation, is the funded account for scheduled and emergent repair work on Navy vessels at both public shipyards and private repair yards. It covers drydockings, restricted availabilities, continuous maintenance, and the overhauls that keep a hull in class.

FACT: the Department of the Navy submitted its FY27 budget request to Congress on April 3, 2026, requesting $14.3 billion in account 1B4B, per the FY27 OMN justification book. The FY26 level was $13.5 billion. FACT: Operation Epic Fury, the sustained US naval campaign against Iran, was already under way when the budget was submitted. FACT: Chief of Naval Operations Adm. Daryl Caudle told the House Armed Services Committee that the FY26 budget "didn't bake in Epic Fury" and that routine operations — maintenance included — have felt the impact. FACT: CNO Caudle stated in August 2026 that the Navy needs between $6 and $8 billion above its FY26 appropriation to be "whole and solvent," per Navy Times reporting on August 28. FACT: a June 2026 White House supplemental request sought $67.1 billion for munitions replenishment and Iran conflict costs, separate from the regular FY27 submission.

The 5.5 percent growth is not a proxy for 5.5 percent more throughput.

The $14.3 billion FY27 request is the pre-planned trajectory, written and submitted while Epic Fury's full cost was still accumulating. It does not obviously include a catch-up component for FY26 maintenance that was deferred when the accounts were drawn down. The supplemental covers munitions and war costs; the maintenance hole from extended operational tempo is a separate question Congress has not yet funded.

The second issue is per-hull scope and when it lands. More than two dozen ships supported Epic Fury operations, per Navy Times. Hulls that ran extended blockade and strike cycles return with operating hours, systems wear, and material degradation beyond what a normal deployment schedule produces. But the more specific problem is interval compression: a ship whose planned availability was scheduled for month twelve of a deployment cycle that instead ran eighteen months arrives with maintenance items from multiple planned intervals all past due simultaneously. Work that was written to be staggered across two or three availabilities now has to execute in one window. The aggregate labor-hours may not be dramatically larger; the peak trade demand within a fixed dock period is. A yard that has staffed against the distributed version of that work cannot execute the compressed version with the same crew.

Assessment: the dollar budget grew 5.5 percent; the per-hull demand on ships that ran extended operational cycles is likely larger than historical scope assumptions, and the timing of that demand is compressed in a way the national budget line does not show. A 5.5 percent proportional growth applied to a yard's FY26 execution number produces a planning figure that is too low on scope, too optimistic on schedule, and wrong about the trade-staffing peak it implies.

Bottom line: look at the specific hull classes in your FY27 availability schedule and identify which were deployed during Epic Fury. For those hulls, plan scope against what the operational cycle actually consumed rather than against the historical availability baseline. The funded account is $14.3 billion; how much of that demand lands at your facility, and at what intensity, depends on which hulls you are scheduled to receive — not on the national growth rate.

Sources: DoN FY27 Budget Request press release, Apr 21, 2026; FY27 OMN Justification Book; Navy Times, CNO $6-8B shortfall, Aug 28, 2026; CNN, Iran war maintenance delays, May 27, 2026 {.sources}

Automation is compressing the machinist trade to a 30-day training curve — and the defense industrial base has seen this film before

Hadrian opened a 2.2 million square-foot manufacturing facility in Cherokee, Alabama in March 2026, backed by $900 million in Navy funding and $1.5 billion in private capital. The ostensible purpose is submarine components. The more important question is what the technology running the facility does to the machinist trade over the next fifteen years.

Hadrian's software platform, called Opus, does not assist machinists. It absorbs the decision-making that previously lived in them. FACT: 256 Today's facility reporting describes Opus as the system that "takes a design file, determines how to manufacture the part, programs the machines, manages the workflow, and inspects the finished product." Navy Secretary Phelan described the model at the opening: "Hadrian does not just machine parts. They build integrated production systems: raw material in, test-ready hardware out." FACT: the facility reports machine uptime of 75 to 80 percent against an approximately 30 percent baseline in traditional precision manufacturing, per the same reporting. FACT: Hadrian

CEO Chris Power stated that Opus enables workers to reach full productivity in 30 to 40 days, against years of apprenticeship in a conventional machine shop. FACT: Hadrian CEO Chris Power told Defense One that the skilled-worker shortage is “the number-one problem” and that automation is the primary path out of it. VAdm. Robert Gaucher, the Navy’s submarine production czar, characterized the challenge as a “math problem” requiring gains across workforce, productivity, and parts delivery, per the same reporting.

This is the same model article one of this issue describes in welding and surface preparation. HII’s $900 million HYPR agreement with Path Robotics and GrayMatter covers autonomous welding, surface prep, and coating across every major HII program. Three high-volume trades — welding, surface preparation, and precision machining — are all receiving technology investment in the same year that absorbs the expert decision into software and reduces the human role to supervision. The production logic is clear and the funding is real.

The question is what happens when the system goes down and nobody can do it by hand.

FACT: in 1983, engineer Lisanne Bainbridge published “Ironies of Automation” in the journal Automatica, a paper that has become foundational in human factors research. Her finding: automation is most likely to fail in the hardest situations — and those are exactly the situations where the human operator, who has been monitoring rather than doing, has had the least practice. The operator is expected to take over. But the skill to take over has atrophied because practice is how skill is maintained, and the automation removed the practice.

Aviation documented the real consequences over the following four decades. Assessment: accident investigators and human-factors researchers have attributed loss of situational awareness and reduced manual proficiency to extended reliance on automated cockpit systems, a pattern consistent with what Bainbridge described. The FAA and major carriers responded with a standing policy: mandatory periodic manual flight practice regardless of automation capability. That policy exists because monitoring automated systems long enough erodes the spatial awareness and fine motor control that manual recovery under stress requires, per Flight Safety Foundation research on automation and manual flying skill.

FACT: the Center for Strategic and International Studies, in its assessment of industrial mobilization and surge capacity, found that the U.S. defense industrial base has become more brittle over time and that specialized skills atrophy without continuous production, with experienced workers migrating when demand drops. FACT: the Army’s own munitions industrial base study found that it takes two years to bring an average line worker to effectiveness, and seven years for specialists in energetics — the explosive propellants and materials that munitions require — and that is in a sector that never fully automated away the underlying trade.

There is a second question the article cannot close. NAVSEA requires specific qualifications for work on Navy vessels — certified welders, qualified inspectors, traceable procedures. Whether Opus-produced components and HYPR-automated welds satisfy those qualification requirements the same way a certified tradesperson’s work does is not addressed in any public source this desk has found. Unverified (operator 2026-09-04): the qualification pathway for automated production within NAVSEA certification frameworks. If the automated system’s output does not carry the same qualification chain as a certified tradesperson’s work, the brittleness problem arrives sooner than the disruption scenario — it arrives the first time a NAVSEA quality representative asks who certified the weld.

Assessment: the Bainbridge irony applies directly to what Opus and HYPR are building. A person trained in 30 days to supervise Opus does not develop the skill to set up a manual lathe, read a machining program from scratch, or establish tolerances by feel. If the Hadrian facility is disrupted — by cyberattack, by supply chain failure on the sensors and chips the platform depends on, or by physical damage — the people inside it cannot do the work by hand. At least pilots fly every day and have natural opportunities to practice the manual skill. A manufacturing supervisor whose job is watching Opus execute has no equivalent moment. The Navy has a policy for maintaining manual flight proficiency. No equivalent policy is known to exist for maintaining manual trade proficiency alongside automated manufacturing. Unverified (no DoD policy document located, operator 2026-09-04): whether any existing program formally requires maintenance of manual machining or welding capability alongside the automated systems now being funded.

> **Bottom line:** the automation investment is justified by today’s production problem. The resilience risk it creates is a fifteen-year problem that nobody is yet formally managing. Ask your workforce development lead whether your training pipeline includes a pathway for maintaining foundational trade skill alongside whatever automation your facility adopts — not to compete with the automated system, but to have the capability when the system is unavailable and the work still has to get done.

Sources: Defense One, Mar 2026; 256 Today, Factory 4 deep dive, Mar 2026; Hadrian press release, Mar 20, 2026; Bainbridge, “Ironies of Automation,” Automatica, 1983; Flight Safety Foundation, “Use It or Lose It”; CSIS, “Industrial Mobilization: Assessing Surge Capabilities, Wartime Risk, and System Brittleness”; Army Science Board, Surge Capacity in the Defense Munitions Industrial Base, 2023 {.sources}

The Waterfront Brief is compiled weekly by the B2 vault's research engine from primary sources and verified trade reporting. Scope: US shipbuilding, US Navy sustainment, and partners across the Indo-Pacific region; out-of-area items only when there is a direct, stated lesson. Freshness policy: every article's anchor source is under 12 months old; older material appears only as dated context or a flagged exception. FACT means the named source says it, not that this desk has independently verified it. Corrections welcome; staged sources are retained for audit.

UPCOMING EVENTS

Nearest first, filling the page. **Bold** marks a decision that closes inside the next 60 days. ASNE is the American Society of Naval Engineers; NSRP is the National Shipbuilding Research Program, the industry-Navy body that funds shared shipyard process research; SCA is the Shipbuilders Council of America.

Dates	Event	Register by	Why it matters
Sep 22–24, 2026	Fleet Maintenance & Modernization Symposium (FMMS), Virginia Beach Convention Center, Virginia Beach, VA (ASNE)	Standard rate now open	The flagship fleet maintenance event for this readership, co-located with the NSRP panel meeting and the Port Engineers Symposium
Sep 29 – Oct 2, 2026	SCA Fall Membership Meeting, Washington, DC (Shipbuilders Council of America)	Members only; no deadline published	The surface repair industrial base's primary Hill-and-Navy engagement window
Oct 28–29, 2026	Arctic Operations Symposium, Maritime Institute of Technology, Baltimore, MD (ASNE)	Not published yet	Cold-region hull and machinery sustainment
Nov 16–17, 2026	Navy Procurement Outlook Summit, Carahsoft Conference Center, Reston, VA (Defense Leadership Forum)	Eventbrite open; no deadline	Where the Navy states shipbuilding-plan and procurement priorities for 2027 — earliest public read on work timing
Dec 7–9, 2026	Technology, Systems & Ships / Combat Systems Symposium, George Mason University, Arlington, VA (ASNE)	Not published yet	Technology and systems exchange near the NAVSEA customer
Jan 12–14, 2027	Surface Navy Association National Symposium, Hyatt Regency Crystal City, Arlington, VA (SNA)	Not published yet	Surface combatant readiness and acquisition priorities
Jan 26–28, 2027	Military Additive Manufacturing Summit (MILAM), Tampa, FL (Defense Strategies Institute)	**Next tier closes Oct 2, 2026**	Metal and polymer AM print qualification and NAVSEA-spec adoption
Mar 23–25, 2027	NSRP All Panel Meeting 2027 (NSRP)	Coming soon per NSRP	Where yard R&D ideas get teamed program-wide
Apr 4–7, 2027	Sea-Air-Space 2027, Gaylord National Resort, National Harbor, MD (Navy League)	Not open yet	The premier US maritime defense exposition — where the Navy states program priorities and procurement direction for the year ahead
Apr 11–15, 2027	AMPP Annual Conference + Expo 2027, Greater Columbus Convention Center, Columbus, OH (AMPP)	Not open yet	The leading corrosion and coatings professional event; preservation is the highest-volume recurring cost on surface ships
Apr 13–15, 2027	RAPID + TCT 2027, Huntington Place, Detroit, MI (SME)	None published	The largest additive manufacturing show in North America
Jun 7–9, 2027	MegaRust 2027, Hampton Roads Convention Center, Hampton, VA (ASNE)	Not open yet	Navy corrosion event; preservation is a top waterfront cost driver